

Orbit Determination System with Reasonable Performance Using Low-Cost Ground Station for Nanosatellite Projects

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Recently the activities for the development of micro- and nano- satellites like a cubesat are being increased. A lot of satellites are operated at a ground station located in a university, but public orbit information is used. In this paper, the orbit determination system is constructed using observations of Doppler frequency at a low-cost ground station. The traditional batch state estimation filter is used, and the orbital elements of satellite position and velocity in inertia coordinates are determined from the Doppler frequency of receiving signals. The verification tests are conducted using UHF-band signals (about 430 MHz) received at an amateur radio station and S-band signals (about 2.2 GHz) received at a 2.4-m parabola antenna from real satellites, and the valid performance is conformed for tracking satellites not depending on public orbit information. The error is max. 0.3 degrees in direction and max. 3.9 km in position for UHF-band signals, and max. 1.1 degrees in direction and max. 37 km in position for S-band signals. This method is valid especially for low-earth-orbit satellites with large Doppler effect.

Key Words: Orbit Determination, Low-Cost Ground Station, Nanosatellite, Doppler Frequency

1. Introduction

Recently the activities for the development of micro- and nano- satellites like a cubesat are being increased. A lot of satellites are operated at a ground station located in a university, but public orbit information released by NORAD (North American Aerospace Defense Command) are used. However, some satellites could not be sufficiently tracked even if public orbit information are used such as a microsatellite UNITEC-1 (UNISEC technological experiment carrier-1) launched in May 2010 which was developed by Japanese universities and small companies (UNISEC), and a nanosatellite KSAT (Kagoshima Satellite) launched together which was developed by Kagoshima Univ. In this paper, the orbit determination system is constructed using observations of Doppler frequency at a low-cost ground station shown in Fig.1. The verification tests are conducted using UHF-band signals received at an amateur radio station and S-band signals received at a 2.4-m parabola antenna from real satellites, and the valid performance is conformed for tracking satellites not depending on public orbit information.

The NORAD releases orbit information on the WEB in several hours after a satellite is launched. In the case of a satellite in sun synchronous orbit at 700-km altitude, it is often updated every half day. However, in the operation of UNITEC-1 which is a deep space spacecraft to Venus with the primary satellite Akatuki, the operating university was responsible for the orbit determination. In the operation of KSAT which altitude is less than 300 km, the update of orbit by NORAD was not stable and it was sometimes stopped for four days. The tracking telemetry signals were very

difficult because the orbit was often changed by strong atmospheric drag.

In the future, it is important subject to prevent space debris increasing, so nanosatellites launched in an orbit at less than 300-km altitude will be increased because the orbit can be naturally decreased in less than 1 year. It is very important that the system of orbit determination and tracking satellites not depending on NORAD is prepared in satellite developing organizations.

The orbit determination method using Doppler frequency and batch state estimation filter was suggested in about 50 years ago¹⁾. In real projects, the measurements of range and range rate (RARR) have been traditionally used²⁾, and the accuracy is typically 15

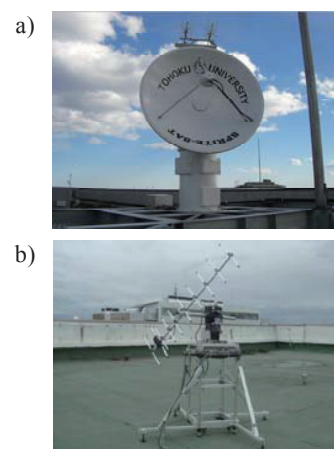


Fig. 1. Satellite tracking antennas at Tohoku Univ., Sendai JP: a) 2.4-m parabola antenna for S-band downlink, b) 10-element Yagi antenna for 435MHz downlink.

m and 1 mm/s. In other methods, the measurements using satellite laser ranging (SLR) systems enable us to observe the range with an accuracy of about ± 1 cm, but the number of ground stations is not sufficient. The direct position measurements with on-board GPS receivers are easily utilized for microsatellites with an accuracy of 10 meters, but some nano satellites do not have a GPS receiver because of saving power consumption.

The methodology of orbit determination in this paper is not original, but the real observations and useful performance evaluation using an existing university low-cost ground station are not mentioned in other researches. In this system, it is not required a special onboard instrument for satellites, and the measurements of Doppler frequency can be obtained from all the LEO satellites. It is the merit that many universities and institutions can easily adopt and try this method using an existing ground station conventionally used for only data communication.

2. Theory

When satellite signals are received at ground, the distance or range ρ between a satellite and a ground station is gradually changed, and the radio frequency has Doppler shift. In the inertial coordinate system, the position and velocity vectors of a satellite and a ground station can be calculated by orbital theory. From the relative position and velocity vectors $\mathbf{p}$, $\dot{\mathbf{p}}$, the range-rate $\dot{\rho}$ which is change rate of range to time can be derived by

$$\dot{\rho} = \mathbf{p} \cdot \dot{\mathbf{p}} / \rho \quad (1)$$

The carrier frequency of telemetry signals is defined by f_0 , and light speed is defined by c , the Doppler frequency f can be defined by

$$f = (1 - \dot{\rho}/c)f_0 \quad (2)$$

The orbit is defined by a state vector $\mathbf{X}_0 = \{x, y, z, dx/dt, dy/dt, dz/dt, f_0\}^T$ at an epoch time t_0 . Where, each element is position or velocity in earth-centered equatorial inertia coordinate frame, and the carrier frequency f_0 can be also included as an estimated parameter. The group of measurement f_i at time t_i is defined as observation vector $\mathbf{Z}$. In the orbit determination, $\mathbf{X}_0$ is updated to decrease the residuals between estimated $\bar{\mathbf{Z}}$ and real $\mathbf{Z}$. The orbit $\mathbf{X}_i$ at time t_i and estimated measurement $\bar{f}_i$ is calculated from the initial $\mathbf{X}_0$. The $\mathbf{X}_i$ can be calculated by the numerical integration method using derivative equations of orbital dynamics model. In the experiments of this paper, simple orbital model is used including only J_2 term. From the residual between estimated $\bar{f}_i$ and real f_i , the $\mathbf{X}_0$ is recursively updated. At final, when the RMS (root mean square) of residuals is converged, the final value is defined by an optimal orbit $\hat{\mathbf{X}}$. These sequences are the application of general state estimation filter.

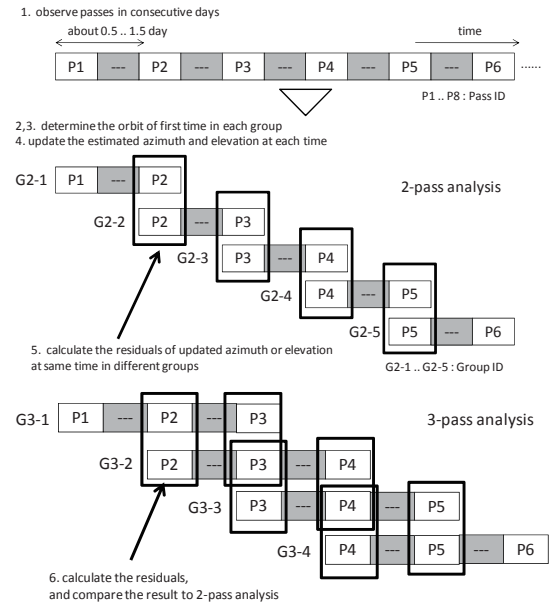


Fig. 2. Evaluation procedure of orbit determination.

Adding small changes to $\mathbf{X}_0$ such as 1m or 1mm/s, the relation between small change $\Delta\mathbf{Z}$ and $\Delta\mathbf{X}_0$ can be derived by

$$\Delta\mathbf{Z} = H\Delta\mathbf{X}_0 \quad (3)$$

where H is a sensitivity matrix, and $H = \partial\bar{\mathbf{Z}} / \partial\mathbf{X}_0$. This matrix is numerically calculated and updated when $\mathbf{X}_0$ is changed. The error covariance matrix R is defined, where its diagonal elements are observation variance σ^2 and other elements are all zero. Finally, $\Delta\mathbf{X}_0$ can be calculated from $\Delta\mathbf{Z}$ like the following formula. The $\Delta\mathbf{Z}$ is a residual between $\mathbf{Z}$ and $\bar{\mathbf{Z}}$ ($\Delta\mathbf{Z} = \mathbf{Z} - \bar{\mathbf{Z}}$). The $\mathbf{X}_0$ is replaced by $\mathbf{X}_0 + \Delta\mathbf{X}_0$, and this sequence is repeatedly calculated until $\Delta\mathbf{X}_0$ reaches to zero.

$$\Delta\mathbf{X}_0 = (H^T R^{-1} H)^{-1} H^T R^{-1} \Delta\mathbf{Z} \quad (4)$$

The program ODEF (Orbit DEtermination Filter) was developed and used for real orbit observations shown in later sections, which can solve the optimal orbit from the observations of Doppler frequency.

When a target satellite is in low-earth orbit at about 600 to 800-km altitude, the signals can be received at a ground station about 10 to 15 minutes per round. In experiments of this paper, the several patterns of a number of passes simultaneously processed in the program are defined such as two-, three-, four-, or five-passes. The precision of optimal orbit is evaluated in each pattern. For example, the following procedure is in the case of two-pass analysis using total 6 passes. The procedure is also shown in Fig.2.

1. Each pass is numbered as P1 to P6. The each observation time in pass p is defined as $t_{p,i}$ ($i = 1, \dots, N_p$), where N_p is a number of observations in pass p .
2. The analysis groups G2-1 to G2-5 are made from sequential two pass, and observation vector in each group is defined by $\mathbf{Z}_1$ to $\mathbf{Z}_5$.

3. The orbit $\mathbf{X}_1$ to $\mathbf{X}_5$ at the first observation time in each group is updated to optimal orbit $\hat{\mathbf{X}}_1$ to $\hat{\mathbf{X}}_5$ by the program ODEF.
4. The azimuth α and elevation δ of target satellite at ground station are calculated at time $t_{p,i}$. The values calculated by orbit $\hat{\mathbf{X}}_k$ is defined by $\alpha_{p,i}(\hat{\mathbf{X}}_k)$ or $\delta_{p,i}(\hat{\mathbf{X}}_k)$.
5. The each group has some observations at same time. The residuals $\Delta\alpha$ or $\Delta\delta$ are calculated from α or δ of different groups. The RMS of residuals are defined by σ_α or σ_δ . The following formula is σ_α , but σ_δ can be calculated by replacing α to δ . Where, the values at more than 80-deg elevation are omitted.

$$\begin{aligned} \Delta\alpha_{2,1,i} &= \alpha_{2,i}(\hat{\mathbf{X}}_1) - \alpha_{2,i}(\hat{\mathbf{X}}_2) \\ \dots \quad \Delta\alpha_{5,4,i} &= \alpha_{5,i}(\hat{\mathbf{X}}_4) - \alpha_{5,i}(\hat{\mathbf{X}}_5) \end{aligned} \quad (5)$$

$$\sigma_\alpha = \sqrt{\left\{ \sum_{k=1}^4 \sum_{i=1}^{N_p} (\Delta\alpha_{(k+1),k,i})^2 \right\} / \sum_{p=2}^5 N_p} \quad (6)$$

6. When total pass number is defined by M_g , and number of passes in each group is defined by N_g , the general formula is defined as follows. In the example of procedure 1 to 5 is in the case of $M_g = 6$, $N_g = 2$.

$$\Delta\alpha_{p,k,i} = \alpha_{p,i}(\hat{\mathbf{X}}_k) - \alpha_{p,i}(\hat{\mathbf{X}}_{k+1}) \quad (7)$$

$$a = N_g + k - 1, \quad b = M_g - N_g \quad (8)$$

$$\sigma_\alpha = \sqrt{\frac{\sum_{k=1}^b \sum_{p=k+1}^a \sum_{i=1}^{N_p} (\Delta\alpha_{p,k,i})^2}{\sum_{k=1}^b \sum_{p=k+1}^a N_p}} \quad (9)$$

3. Experiment A: Observation of UHF Signals

3.1. Measurement equipment

The simple receiving antenna is fabricated shown in Fig.1-b). It consists of a 10-element Yagi antenna (A430S10R by Diamond Antenna Corp.) with 11.5dBi gain and a tracking rotator (G-5500 by YAESU Corp.).

At the zero elevation, the height of tower is 1.2 m, and the width is 1.5 m. It can stand up to 50-m/s winds theoretically.

The telemetry signals are received using 140/435-MHz amateur radio transceiver (IC-910D by ICOM Corp.). The method was suggested by Oda et al.^{3,4)} using the output voice signals from transceiver. Recording signals by a sound card with PC, the frequency is analyzed using general FFT software. Then the value of offset frequency to the setting value of transceiver can be measured. The change rate of voice frequency and the one of radio frequency are equivalent. Using this method, a expensive spectrum analyzer is not necessary to measure the value of radio frequency.

The frequency stability of transceiver was 3×10^{-6} originally. This means that the frequency deviation of 1308 Hz at maximum to 436-MHz signals happens under the several temperatures or time passing. This performance is not sufficient for orbit determination, so the reference 10-MHz signal generated by an OCXO

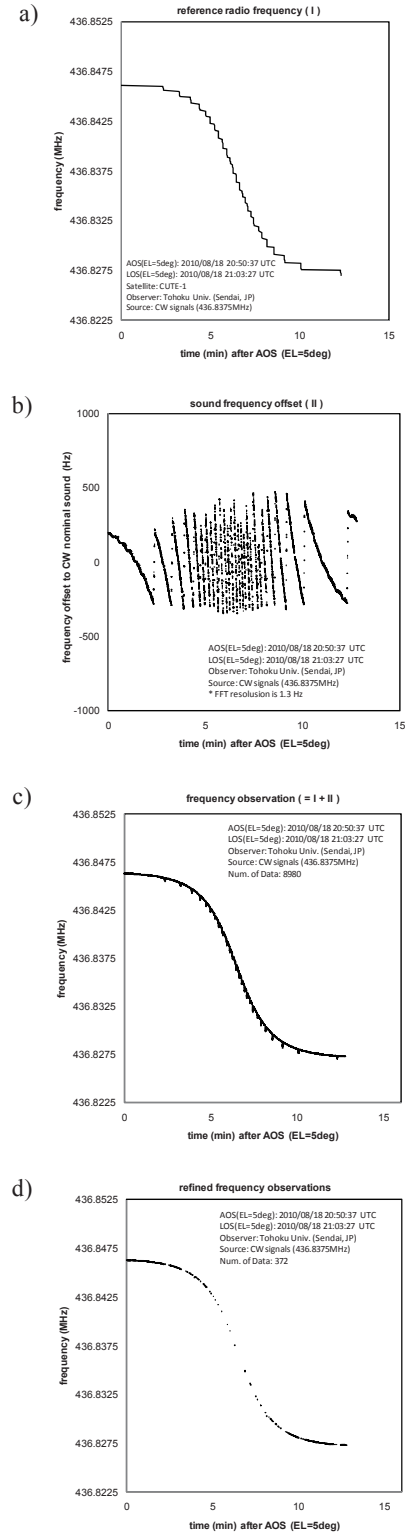


Fig. 3. Pre-processing of observation data: a) setting frequency of receiver, b) frequency of CW sound, c) sum. of setting and sound freq., d) refined data after filtering.

(oven controlled xtal oscillator) with a stability of 1×10^{-8} is inputted to the transceiver. When the 436-MHz signals are measured by this transceiver for 10 minutes

Table 1. List of observation passes (CUTE-1).

Pass ID	AOS JST*	LOS JST	Span per pass min	Span per pass sec	Total Span day	Max. EL deg	Num. of Data**
P1	2010/12/16 17:39:13	17:51:33	12	20	0.0	48.0	275
P2	2010/12/17 05:58:09	06:10:59	12	50	0.5	78.4	337
P3	2010/12/18 05:38:16	05:50:46	12	30	1.5	50.6	318
P4	2010/12/19 16:39:23	16:51:43	12	20	3.0	46.6	234
P5	2010/12/20 17:59:53	18:11:33	11	40	4.0	30.9	224
P6	2010/12/21 17:39:27	17:51:47	12	20	5.0	47.8	269
P7	2010/12/22 17:19:16	17:31:56	12	40	6.0	74.6	327
P8	2010/12/23 16:59:19	17:11:59	12	40	7.0	72.4	316
Total							2300

*JST = UTC + 9:00
 ** num. of refined data after filtering

under the constant temperature, the deviation was improved to less than 0.1 Hz with external 10-MHz reference signals, compared to the 5-Hz deviation without it.

The setting frequency to transceiver is not automatically controlled, and it is changed manually when CW voice sound is being listened. The CW voice sound is 800 Hz when the radio frequency is the same as the setting frequency to instrument. The radio frequency is gradually decreased by Doppler effect, so the setting frequency is changed down about 900 Hz when the voice sound reaches to 300 Hz. The setting frequency is automatically recorded by PC, and the radio frequency is the sum of the setting frequency and the voice frequency. For the FFT analysis, the freeware of Spectran (Version 2.0, Special UNITEC-1 Version) developed by Mr. Alberto is used. By this program, the peak frequency and the signal level can be automatically recorded in which the resolution setting is 1.3 Hz.

3.2. Pre-processing of observation data

The target satellite is the cubesat CUTE-1 (Sat. ID: 27844) developed by Tokyo Institute of Technology. The size is 10 cm each side, the weight is about 1 kg, and the orbit is 820-km-alt. sun synchronous. The operation is still continued at the time of April 2010 though it was launched in June 2003. The CW beacon signals with 436.8375 MHz and 0.1 Watt are measured.

The one example of frequency measurement is shown in Fig.3. The a) is history of setting frequency to transceiver, and b) is the CW sound frequency after the FFT analysis. The setting frequency is controlled manually, so the line of a) is not smooth. The values in b) are deviation to 800-Hz nominal frequency. The c) is the sum of the setting frequency and the sound frequency. The number of data of c) is very large, so the filtering procedure is applied to decrease the number of data and omit some data with large noise.

1. Erase data with low signal level less than a threshold value
2. Erase data measured when the elevation is less than 5 degrees
3. Reduce the number of data by calculating the average value every one second
4. Erase data with more than 10-Hz error to the approx. line generated by least squares method every 30 seconds

3.3. Result of orbit determination

In the 7-day span from Dec. 16, 2010 to Dec. 23, 2010,

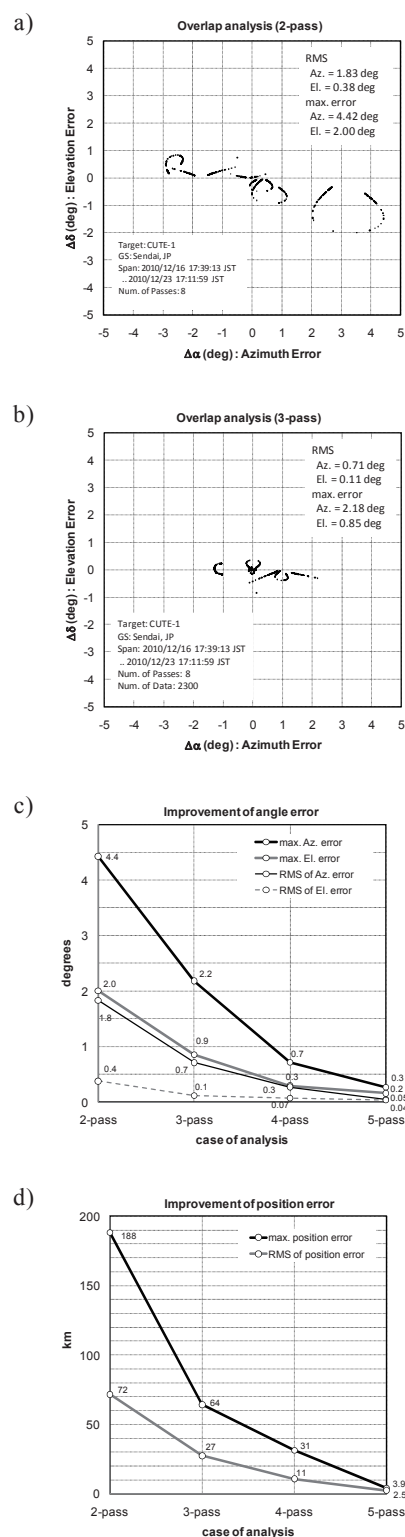


Fig. 4. Result of orbit determination for CUTE-1: a) angle errors by 2-pass analysis, b) angle errors by 3-pass analysis, c) improvement of angle errors by 5-pass analysis, d) improvement of position errors by 5-pass analysis.

total 8 passes were observed shown in Table 1. Firstly, the program ODEF is used for only single pass, and the

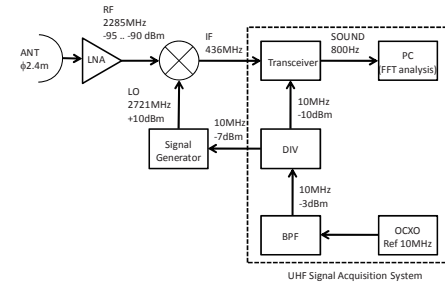


Fig. 5. Signal acquisition system for S-band downlink.

Table 2. List of observation passes (SPRITE-SAT).

Pass ID	AOS JST*	LOS JST	Span per pass min sec	Total Span day	Max. EL deg	Num. of Data**
P1	2011/01/14 00:16:47	00:27:37	10 50	0.0	49.6	68
P2	2011/01/14 12:12:07	12:22:37	10 30	0.5	38.6	28
P3	2011/01/15 12:42:59	12:53:59	11 0	1.5	85.7	124
P4	2011/01/16 13:14:10	13:24:50	10 40	2.5	42.6	68
P5	2011/01/17 12:08:21	12:18:41	10 20	3.5	35.0	53
P6	2011/01/18 12:39:10	12:50:10	11 0	4.5	78.6	116
Total						457

*JST = UTC + 9:00

** num. of refined data after filtering

parameter of carrier frequency f_0 was determined for each pass. The average offset value is -1676 Hz to the nominal value of 436.8375 MHz in which the max. deviation is 33 Hz and the standard deviation is 17.6 Hz. The determined values different in each pass are applied to observation model, and the carrier frequency f_0 is removed from state vector.

Secondly, multi-pass analyses are carried out, and the angle error is evaluated by overlap analysis shown in Section 2. In the 2-pass analysis, the max. azimuth error is 4.42 degrees and the RMS is 1.83 degrees shown in Fig.4-a). In the 3-pass analysis, the cloud of angle errors shown in Fig.4-b) is smaller than 2-pass analysis, the max. azimuth error is 2.18 degrees and the RMS is 0.71 degrees. After the 3-pass/4-pass/5-pass analyses, the history of angle errors is summarized in Fig.4-c). In the 5-pass analysis, the azimuth error is decreased to 0.26 degrees in max. error, and 0.05 degrees in RMS. Also, the position errors are summarized in Fig.4-d). In the 5-pass analysis, the error is decreased to 3.9 km. From these results, it was proved that this hardware system and analysis method is valid to real satellite observations, and it was confirmed that the precisions of orbit determination can be achieved less than 0.3 degrees in azimuth, less than 0.2 degrees in elevation, and less than 4 km in position.

4. Experiment B: Observation of S-band Signals

The target satellite is the 45-kg microsatellite SPRITE-SAT (RISING) (Sat. ID: 33494) developed by Tohoku University, which orbit is 660-km-alt. sun synchronous. It was launched on Jan. 23, 2009, and it is transmitting CW signals (2285 MHz, 0.1 Watt) without modulation.

The 2.4-m parabola antenna used in S-band observations has about ± 2.2 degrees of half beam width (-3dB power) for the radio frequency of 2285 MHz.

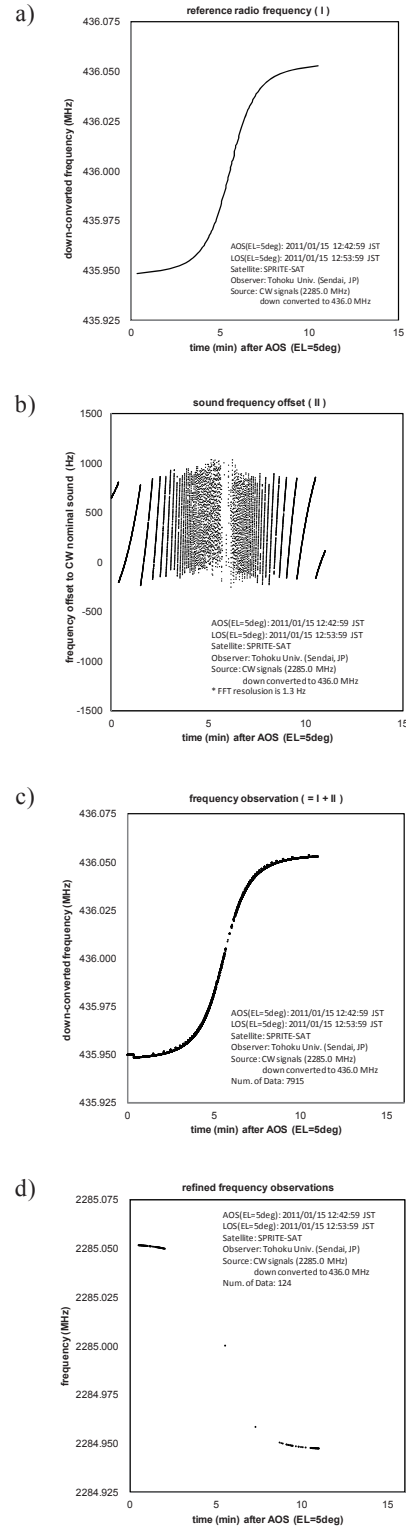


Fig. 6. Pre-processing of SPRITE-SAT observation data: a) setting frequency of receiver, b) frequency of CW sound, c) sum. of setting and sound freq., d) refined data after filtering.

The target precision of orbit determination must be less than this half beam width to continue the tracking satellite without NORAD orbit information.

The signal acquisition system for S-band downlink is shown in Fig.5. The system is extension of U-band system, and the signals of 2285 MHz are down converted to 436 MHz. The signals of 2721 MHz generated by a signal generator (N9310A by Agilent Corp.) are inputted to a mixer. The reference 10-MHz signals with the stability of 1×10^{-8} are used to both of transceiver and signal generator.

The sum of setting frequency and sound frequency is defined by IF frequency f_{IF} , where $f = f_{L.O} - f_{IF}$. For the program ODEF, the radio frequency f is applied.

In the 5-day span from Jan. 14, 2011 to Jan. 18, 2011, total 6 passes were observed shown in Table 2. The pre-processing of observation data for Pass-3 as an example is shown in Fig.6. Firstly, the carrier frequency f_0 was determined and the average offset value is - 218 Hz to the nominal value of 2285 MHz in which the max. deviation is 487 Hz and the standard deviation is 245 Hz. The frequency stability is same as the observations of CUTE-1, so the order of errors is almost 10 times to the result of CUTE-1 analysis.

Secondly, multi-pass analyses are carried out. In the 2-pass/3-pass analysis, angle errors of all the observations are plotted in Fig.7-a) and Fig.7-b), the max. and RMS of angle errors are shown in Fig.7-c), and the max. and RMS of position errors are shown in Fig.7-d). It was confirmed that the precisions of orbit determination can be achieved less than 0.9 degrees in azimuth, less than 1.1 degrees in elevation and less than 37 km in position from the 3-pass analysis. The angle errors are less than half beam width, and the continuous satellite tracking without NORAD orbit can be expected.

5. Conclusion

In this paper, the low-cost orbit determination system was suggested. The utilization of system is proved by the real observations and analyses for U-band radio signals of CUTE-1 and S-band signals of SPRITE-SAT. The achieved precision of orbit determination was 0.3 degrees in azimuth, 0.2 degrees in elevation, and 4 km in position in max. error from the 7-day CUTE-1 observations, and it was 0.9 degrees in azimuth, 1.1 degrees in elevation, and 37 km in position in max. error from the 5-day SPRITE-SAT observations.

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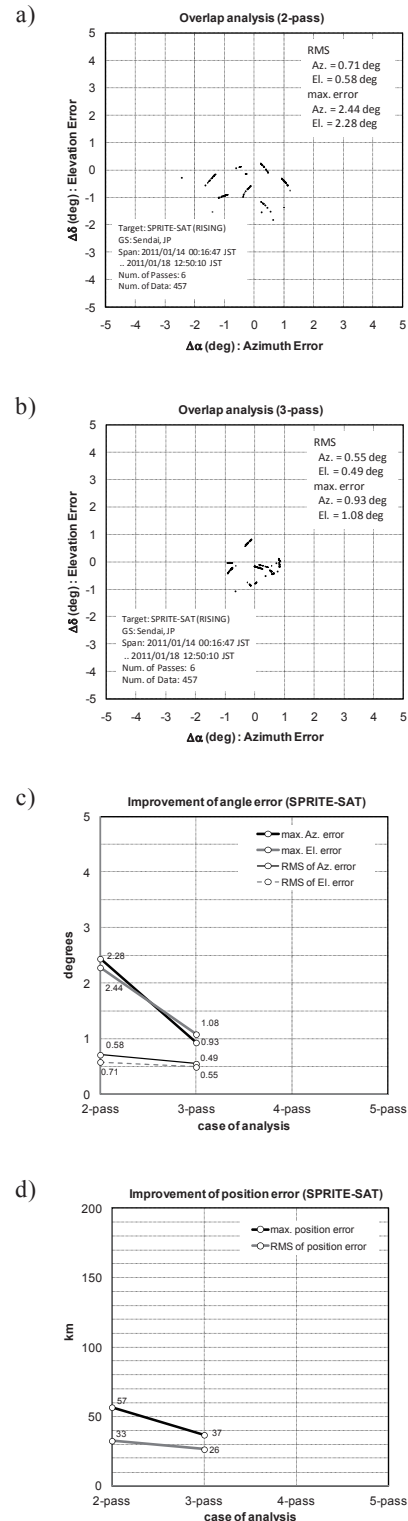


Fig. 7. Result of orbit determination for SPRITE-SAT: a) angle error of 2-pass analysis, b) angle error of 3-pass analysis, c) improvement of angle errors by 3-pass analysis, d) improvement of position errors by 3-pass analysis.